

SCPN Control: Neuro-Symbolic Stochastic Petri Net Controllers with First-Principles Gyrokinetic Transport for Real-Time Tokamak Plasma Control

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17 March 2026

Summary

`scpn-control` is an open-source neuro-symbolic control engine for tokamak plasma control combining Stochastic Petri Net (SPN) compilation into spiking neural network (SNN) controllers, a five-tier gyrokinetic transport system including a nonlinear δf solver, and an 8-layer Kuramoto-Sakaguchi phase dynamics engine.

The electromagnetic gyrokinetic and Lyapunov-monitor surfaces are repository research-prototype evidence, not facility-control guarantees. The finite-beta KBM/MTM branch logic has regression coverage, but nonlinear electromagnetic extension evidence and external-code or facility validation remain open. The Lyapunov guard is an advisory sliding-window monitor: it can approve during warm-up and until the configured consecutive-violation threshold is exceeded, so it is not a standalone fail-closed safety interlock.

The package provides: a Grad-Shafranov equilibrium solver (fixed and free boundary, JAX-differentiable), a 1.5D Crank-Nicolson transport solver with multi-ion physics, a native linear gyrokinetic eigenvalue solver with electromagnetic extension, a native TGLF-equivalent quasilinear model (SAT0/SAT1/SAT2 saturation rules), a nonlinear δf gyrokinetic solver in flux-tube geometry with a JAX GPU path, interfaces to five external GK codes (TGLF, GENE, GS2, CGYRO, QuaLiKiz) whose quantitative external-code claims remain blocked until real artefacts are admitted, a hybrid surrogate+GK validation layer with out-of-distribution detection and online retraining, ten controllers (PID, MPC, H_∞ , Riccati state feedback with static μ analysis, NMPC, SNN, safe RL, sliding-mode, gain-scheduled, fault-tolerant), disruption prediction with SPI mitigation, and a companion Rust backend (5 crates, PyO3 bindings) with a reproducible benchmark reporting a $\sim 5 \mu\text{s}$ P50 native integrated control cycle on the CI runner ($2.85 \mu\text{s}$ on the local workstation).

The nonlinear gyrokinetic solver implements the Cyclone Base Case configuration (Dimits et al. 2000), Rosenbluth-Hinton zonal-flow damping, kinetic electrons, and Sugama collisions. After the v0.19.0 physics audit, its late-time nonlinear heat-flux normalisation is treated as an open validation target: the latest 2000-step adiabatic run had not reached saturated χ_i , and the linear local-dispersion path overpredicts the published GENE growth rate. The manuscript therefore does not claim quantitative nonlinear CBC agreement until a longer, reverified convergence campaign is complete.

The codebase comprises a Python package and 5 Rust crates with broad Python module tests, a 100% local package-coverage gate configured consistently for CI, and a multi-workflow CI matrix.

Statement of Need

Real-time plasma control in magnetic confinement fusion requires low-latency decision paths, bounded and traceable transport-physics evidence, and robust safety guarantees. Existing open-source tools address subsets of this problem: TORAX (Citrin, Goodfellow, et al. 2024) provides JAX-differentiable transport with TGLF coupling, TCV-RL (Degraeve et al. 2022) applies deep RL to tokamak control, FreeGS (Dudson 2023) solves free-boundary equilibria, and FUSE (Meneghini, Slendeboek, et al. 2024) offers integrated design-to-operations modelling. These systems establish distinct reference points rather than a single matched benchmark.

This paper evaluates four connected `scpn-control` capabilities:

1. **Five-tier gyrokinetic transport** — spanning critical-gradient models ($\sim \mu\text{s}$), a QLKNN-style neural-transport facade with a compact MLP loader and analytic critical-gradient fallback, a native linear GK eigenvalue solver (approximately 0.3 s per flux surface in the recorded repository context) (Dimits et al. 2000; Miller et al. 1998), a native TGLF-equivalent model with SAT0/SAT1/SAT2 spectral saturation (Staebler, Kinsey, and Waltz 2007; Staebler et al. 2017; Maeyama et al. 2015), and a nonlinear δf gyrokinetic solver with dealiased $\mathbf{E} \times \mathbf{B}$ bracket, ballooning connection boundary conditions, Rosenbluth-Hinton zonal flow physics (Rosenbluth and Hinton 1998), Sugama collision operator (Sugama and Watanabe 2006), and optional kinetic electrons via semi-implicit backward-Euler treatment. The JAX GPU path is not promoted here to a portable speedup or facility timing claim. The current public neural-transport claim report is `analytic_fallback`; quantitative QLKNN/QuaLiKiz claims remain blocked unless trained weights and strict reference artefacts are supplied.
2. **Cyclone Base Case validation harness** — the repository includes CBC scenarios, grid scans, kinetic-electron lanes, and published-code interface adapters. Quantitative nonlinear agreement is not yet claimed: the current audited result is an unsaturated adiabatic heat-flux trace

after 2000 steps, and the linear local-dispersion path overpredicts the GENE reference growth rate. The remaining publication gate is a longer reverified convergence campaign plus native GK comparison against real TGLF/GENE-class binaries on identical inputs.

3. **SPN-to-SNN compilation** — translates control graphs into leaky integrate-and-fire neuron pools with stochastic bitstream encoding (Murata 1989; Maass 1997), enforcing pre/post-condition contracts on every observation and action. The pure-NumPy LIF+NEF engine requires no external dependencies.
4. **8-layer plasma phase dynamics** — a Kuramoto-Sakaguchi multi-layer UPDE engine (Kuramoto 1975; Sotek 2026) encoding experimentally grounded interactions (drift-wave/zonal-flow, NTM/bootstrap-current, ELM/pedestal), with GK-driven adaptive K_{nm} coupling and Lyapunov stability monitoring. The monitor is a research-prototype advisory guard with warm-up and consecutive-window fail-open intervals, not a replacement for an independent facility safety interlock.

Implementation

The Python package is organised into four layers:

- **Core (`scpn_control.core`):** Grad-Shafranov solver (Picard iteration with multigrid or SOR), 1.5D Crank-Nicolson transport, GEQDSK/IMAS I/O, IPB98(y,2) scaling (ITER Physics Basis Editors 1999), neural equilibrium accelerator, uncertainty quantification, and the gyrokinetic transport stack:
 - *Native linear GK*: Miller flux-tube geometry (Miller et al. 1998), per-species Gauss-Legendre velocity grid, local dispersion relation with Newton root-finding, mixing-length quasilinear fluxes. Electromagnetic extension adds KBM (Tang, Connor, and Hastie 1980) and microtearing (Drake and Lee 1977) modes as repository research-prototype evidence pending nonlinear EM and external-code revalidation.
 - *Native TGLF-equivalent*: SAT0/SAT1/SAT2 spectral saturation (Staebler, Kinsey, and Waltz 2007; Staebler et al. 2017), $E \times B$ shear quench (Waltz, Kerbel, and Milovich 1997), trapped-particle damping (Connor and Hastie 1974), multi-scale ITG-ETG coupling (Maeyama et al. 2015). No external binary required.
 - *Nonlinear δf* : 5D Vlasov in flux-tube geometry, dealiased $E \times B$ bracket (Orszag 2/3 rule), 4th-order parallel streaming with ballooning connection BC, Rosenbluth-Hinton zonal Krook damping (Rosenbluth and Hinton 1998), Sugama collision operator (Sugama and Watanabe 2006) with particle/momentum/energy conservation, optional kinetic electrons (semi-implicit backward-Euler for electron parallel streaming), RK4 with CFL-adaptive dt, JAX-accelerated variant with the

- recorded bounded local GPU speedup described above.
- *External GK*: TGLF (Staebler, Kinsey, and Waltz 2007), GENE (Jenko et al. 2000), GS2 (Kotschenreuther, Rewoldt, and Tang 1995), CGYRO (Candy and Waltz 2003), QuaLiKiz (Bourdelle et al. 2007) via subprocess with automatic input deck generation and output parsing.
 - *Hybrid*: OOD detection (Mahalanobis + ensemble + range), spot-check scheduling, multiplicative/additive correction with EMA smoothing, online surrogate retraining with validation holdout and rollback.
 - **Phase** (`scpn_control.phase`): Kuramoto-Sakaguchi stepper, UPDE multi-layer solver, adaptive K_{nm} engine with GK→UPDE bridge, Lyapunov guard, WebSocket real-time monitor.
 - **SCPN** (`scpn_control.scpn`): SPN structure, compiler, contract system, artifact serialisation, FPGA HDL project export evidence.
 - **Control** (`scpn_control.control`): H_∞ (Riccati DARE), Riccati state feedback with a D-scaling upper bound (bounded static μ -analysis, not validated full DK synthesis), NMPC (SQP), gain-scheduled PID, shape controller, safe RL (CPO-formulated Lagrangian constraints with control barrier functions, Ames 2017), sliding-mode vertical stability, scenario scheduler, fault-tolerant control, digital twin, flight simulator, Gymnasium environment, JAX-traceable runtime, ITER CODAC/EPICS interface, federated disruption prediction.

The Rust backend (`scpn-control-rs`, 5 crates, ndarray 0.16, rand 0.9) provides PyO3 bindings for performance-critical paths. Its reproducible benchmark reports a ~ 5 μ s P50 native integrated control cycle on the CI runner (2.85 μ s locally); production runtime claims remain subject to runtime-admission evidence.

Validation

Repository validation evidence includes:

- **Cyclone Base Case** (Dimits et al. 2000): CBC input construction, nonlinear δf evolution, JAX acceleration, kinetic-electron support, Sugama collisions, and grid-scan machinery are implemented. The latest audited 2000-step adiabatic run did not reach saturated χ_i , so the former saturated-heat-flux agreement claims are withdrawn pending a longer convergence campaign. The current linear local-dispersion result overpredicts the GENE reference growth rate; it is retained as a model limitation, not a quantitative validation claim.
- **Dimits shift** (Dimits et al. 2000): the repository contains subcritical and supercritical scan machinery and zonal-flow damping, but the full Dimits-gap claim remains a revalidation target after the kinetic-electron and Gauss-Laguerre quadrature changes.

- **Sugama collision operator:** pitch-angle scattering with energy-dependent collision rate ($\nu(v) \propto v^{-3}$) and conservation corrections. Verified: $\int C[f] dv < 3 \times 10^{-8}$ (particles), $\int v_{\parallel} C[f] dv < 10^{-23}$ (momentum), $\int E C[f] dv < 2 \times 10^{-8}$ (energy). At low collisionality ($\nu = 0.01$), Sugama and Krook agree on the low-collisionality collision response in the verified operator tests; saturated nonlinear χ_i remains part of the CBC revalidation target.
- **SPARC/ITER equilibria:** regression-gated against CFS SPARCPublic GEQDSK files and ITER design parameters; this is not independent same-case experimental reconstruction evidence.
- **DIID-D disruption fixtures:** synthetic scenarios covering multiple disruption classes; they are CI plumbing rather than measured-shot evidence.
- **IMAS round-trip:** real omas ODS for equilibrium and core_profiles IDS.
- **IPB98(y,2):** ITPA 20-tokamak H-mode confinement database (ITER Physics Basis Editors 1999).

The test suite comprises Python module tests and Rust workspace tests across CI jobs (Python 3.11–3.13 on Linux, Python 3.12 on Windows and macOS, Rust stable, JAX parity, CodeQL security analysis, OpenSSF Scorecard). The local and CI coverage configuration currently enforces a 100% package-coverage gate while publishing XML coverage artefacts. Physics equations cite their source papers, and named integration tests exercise cross-module chains such as bootstrap→NTM, EPED→Troyon, L-H→EPED, and runaway→SPI.

Limitations: external GK interfaces are mock-tested (no real Fortran binaries in CI); DIID-D shots use synthetic data, not real MDSplus archives; the neural equilibrium has not been cross-validated against P-EFIT on identical equilibria.

Acknowledgements

We thank the CFS SPARCPublic team for SPARC equilibrium data, the ITPA H-mode confinement database contributors, and the GACODE, GENE, GS2, and QuaLiKiz development teams for their publicly documented input/output formats.

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